



The promise of automated compliance checking

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ARTICLE INFO

Keywords:

BIM
Normative provisions
Compliance checking
Automation

ABSTRACT

The quest to automate compliance checking processes needed for planning, design, construction and operations has been an active research topic for half a century. Literature has provided evidence of extensive research effort, with varying degrees of success, alongside the evolution of computing and supporting technologies. New innovations in theoretical and applied computing, coupled with a strong industry demand, has opened up opportunities that promise to bring this quest closer to reality. This paper reviews evolving approaches for automated compliance checking, addressing challenges in sharing digital architectural and engineering design information, formalising normative provisions as computable rules, and methods of processing them for compliance. The paper identifies current state-of-the-art implementations, discusses challenges faced in national efforts, and highlights future pathways.

1. Introduction

Compliance permeates all activities in the design, construction, and the use of a built environment to ensure it is fit for purpose, constructed in accordance with the design brief, functional and cost-effective to run, safe to use, and sustainable to the environment throughout its service life.

The compliance checking process occurs constantly throughout all phases of a project lifecycle in the AECO (Architecture, Engineering, Construction, and Operation) domain (Fig. 1) and affects all aspects of the lifecycle (Dimyadi and Amor, 2013), which is underpinned by codes and standards:

- The planning and design aspects of a project must comply with, on one hand, the project brief and specifications, which stipulate the client's requirements and the manner in which the project should be executed, and, on the other hand, normative planning, resource, and land use provisions, as well as building design codes and standards.
- The construction phase post-consenting is subject to health and safety provisions, contractual obligations, as well as satisfying consenting conditions, by-laws, and other regulatory constraints.
- Post construction, there are project hand-over procedures to follow, which lead on to facility and asset operations and maintenance requirements that continue to apply until the need to replace or renew an asset, which is also driven by safety, reliability, functional and performance objectives.

There are essentially two main ingredients or data sources in a typical automated compliance checking (ACC) process (Fig. 2), namely:

1. The building model, a digital representation of the design data, which is subject to audit for compliance.
2. Normative knowledge, a computable representation of normative provisions stipulated by codes and standards, which is the baseline for the audit.

Numerous approaches have been reported in the literature on how these two ingredients could be processed for compliance. A common approach is to find common objects and attributes, which is the union of the two data sources. This involves, firstly, identifying how best to access information from each data source (Dimyadi et al., 2016a), and, secondly, mapping objects and attributes of one data source with those in the other (Fig. 3) so that relevant rules can be applied accordingly.

A challenge with this approach is that the building model often contains arbitrary objects and attributes that do not necessarily align with those in the rule set, and vice versa. This has been a primary motivation for research in recent years.

Like most processes that require human experts to interpret natural language normative text for compliance checking, the AECO domain has, traditionally, followed manual practice, which is known to be error-prone, inefficient, and a contributing factor to declining productivity in the domain (Boken and Callaghan, 2009), (Gallaher et al., 2004). A major

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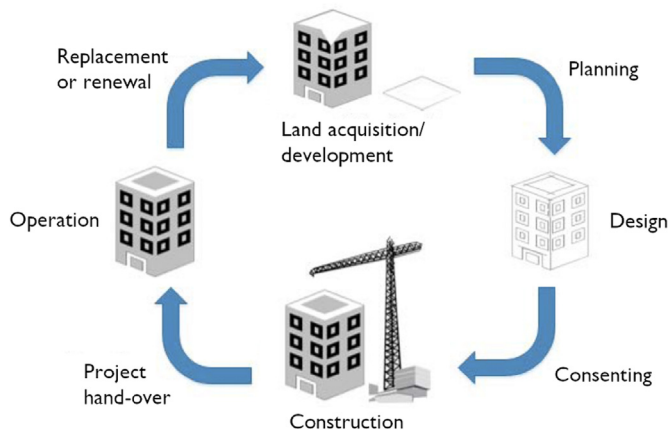


Fig. 1. A typical project lifecycle in the AECO domain.

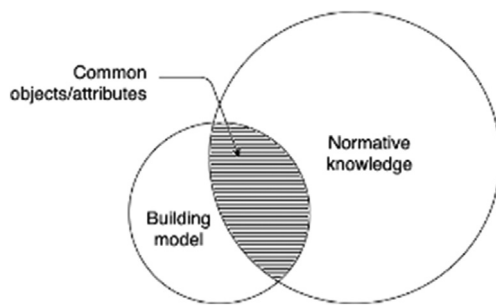


Fig. 2. Core data sources for automated compliance checking.

hurdle to automating the compliance checking process is the challenge in extracting normative knowledge from its natural language source provisions, due to their unsuitability for direct computer processing. For the last 50 years, a common approach is to formalise an interpretation of normative provisions stipulated in building code and standards as computable rules and then embed them into a black-box compliance checking system. This, effectively, creates a digital copy of the paper-based code and standards that is disassociated from the source and must be maintained independently to remain valid. Best practice allows the syntactic and semantic versions to be tightly coupled so any changes in one version can be linked to the related sections of the other version. This is particularly useful when normative knowledge represents regulations or subsidiary legislations that, by nature, are subject to frequent changes.

The emergence of Building Information Modelling (BIM) and the OpenBIM initiative based on the ISO-standard Industry Foundation Classes (IFC) data model (ISO 16739, 2013) in the mid 90's has enabled objects and attributes in the building model to be standardised, although, in practice, the standardisation process poses challenges of its own. The buildingSMART International consortium helped address the issue with the International Framework for Dictionaries (IFD), based on ISO-standard framework for classifications (ISO 12006-3, 2007), (Ekholm, 2005). IFD became the buildingSMART Data Dictionary (bSDD) (Bell and Bjørkhaug, 2006) and has recently been augmented with an API (Application Programming Interface) to empower practitioners to use industry-defined terms and attributes in their building design data set.

On the other side of the object mapping method of ACC, there is still an ongoing quest for an acceptable standard to represent normative provisions for compliance checking purposes (Dimyadi et al., 2017a; Kerrigan and Law, 2003; McGibbney and Kumar, 2013; Solihin et al., 2019). buildingSMART International recently formed the Regulatory Room to address this gap (buildingSMART, 2017). As it stands, however,

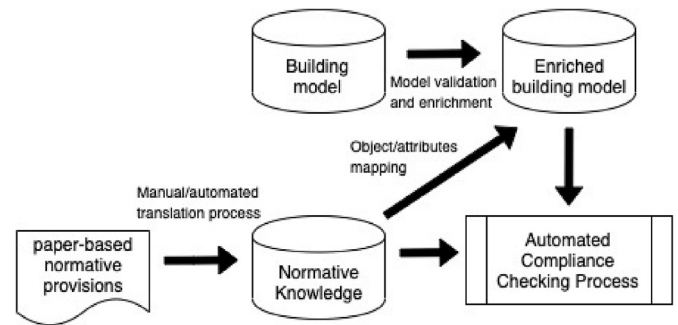


Fig. 3. Typical automated compliance checking process.

the challenge with the object mapping approach remains as there is no guarantee that objects and attributes in any computable rule set can be mapped easily with an industry-standard data dictionary such as the bSDD.

This paper provides an overview of challenges in automating compliance checking processes, summarises different approaches to addressing them as reported in the literature, highlights the most recent progress and identifies future pathways.

2. The known past

The subject of ACC has been an active research topic internationally for more than 50 years (Fig. 4). There are several extensive literature reviews devoted to this topic describing different approaches and systems (Dimyadi and Amor, 2013), (Khemplani, 2018). Some of the systems have since evolved to a higher level of maturity along with the advancement of computing technology and computing applications motivated by the industry's push for innovation and productivity improvements.

2.1. ACC Pre-OpenBIM

Experts systems, or knowledge-based systems, predominated research in AECO as well as the legal domains throughout the 1980's (Eastman, 1981; Hosking et al., 1984, 1987; Maher, 1987; Shaopei and Jinzi, 1987; Sriram et al., 1984; Thomasset and Paquin, 1989). In the early to mid 1990's, the notion of a centralised universal expert system shifted towards distributed knowledge bases on personal computers, providing AECO practitioners with compliance advice on a range of design standards (Amor et al., 1992; Buis et al., 1987; Heikkila and Blewett, 1992; Lutton, 1995).

Prior to OpenBIM, ACC was associated with the ability to share building information (Amor et al., 1990, 1992; Bathurst, 1965; Hosking et al., 1987) in a bespoke form backed by a knowledge-based approach to compliant engineering designs. The earliest successful attempt to represent normative provisions was that of Fenves' Decision Table Formulation (Fenves, 1966), which became the classic example of a successful implementation of Automated Compliance Checking Systems (ACCS) such as SASE, SICAD, and SPEX that could check compliance with the AISC (American Institute of Steel Construction) specifications and served that domain well into the late 1980's (Fenves et al., 1987).

In that period, a few researchers investigated means of capturing and representing normative provisions (Amor, 1992; Feijó et al., 1994; Kähkönen and Björk, 1987; Moulin and Rousseau, 1992; Thomson et al., 1987; Vanier, 1988) for compliant design purposes.

2.2. The OpenBIM approach to ACC

IFC (ISO 16739, 2013) came into being in 1994 and has since facilitated better sharing of building information (Lieblich, 2010) promoted

by the OpenBIM approach, which opens up more opportunities for applications such as model checking and rule-based compliance checking (Kim and Anderson, 2011; Nguyen and Kim, 2011; Taciuc et al., 2016). Several promising ACCS such as DesignCheck in Australia (Ding et al., 2006), ePlanCheck and Fornax in Singapore (Khemlani, 2005), Solibri Model Checker in Finland (Jauhainen, 2010), Jotne EDM in Norway (Yang, 2003), and SMARTCodes in the US (See, 2008), among others, were born around this time. Some have survived until today, but a few have not stood the test of time.

OpenBIM using IFC has been promoted by buildingSMART International as the de facto approach to represent and share rich physical and functional building information among project stakeholders. This has also been the motivation for research to progress towards accessing such complex building data for collaboration and coordination throughout the entire project lifecycle (Cesarotti, 2014; Choi and Kim, 2008; Dimyadi et al., 2016a; Lawrence et al., 2014; Li and Ramani, 2007; Tamke et al., 2014; Zhang et al., 2013).

While OpenBIM provides better building data sharing and collaboration opportunities among project stakeholders, there are currently still some unresolved core issues that hinder the full potential of what automated compliance checking can offer to the industry. These are discussed in Section 3 along with some suggestions for addressing them.

2.2.1. Accessing and sharing OpenBIM data

To minimise the overhead of accessing large data sets from such complex data models as the IFC for specific applications, the ISO 29481 Information Delivery Manual (IDM) and Model View Definitions (MVDs) was adopted by the industry (Gao et al., 2013; Hietanen and Lehtinen, 2014; Jeong et al., 2014; Lee et al., 2016a).

BIM servers such as the open source BIMserver by TNO in The Netherlands also emerged to help with storing, accessing and managing IFC data (Beetz and van Berlo, 2010). A prototype query language plug-in, BIMQL, was developed for this platform to support the need to better access IFC data (Mazairac and Beetz, 2013).

Further efforts in alleviating the difficulty of accessing data IFC coupled with the demand for mapping of objects and attributes across disparate systems sparked the development of linked data and semantic technology by researchers in the AECO domain, coinciding with the development of linked open data in the computing domain in 2007. This initiative has evolved into two active communities, namely W3C Linked Building Data Community Group (<https://www.w3.org/community/lbd/>) and buildingSMART Linked Data Working Group (<https://www.buildingsmart.org/standards/organisation/groups/>). Semantic

technology was based on web technologies such as HTTP, RDF, and URI, intended for both human readability as well as machine processability. A web ontology language (OWL) representation of the IFC, IfcOWL, was developed to enable linking with other data sets that also allows semantic reasoning (Pauwels and Terkaj, 2016), (Beetz et al., 2008). IFC data is available in the RDF (Resource Description Framework) graph data model as directed labelled graphs. Accessing data from IfcOWL has also been made possible through the use of SPARQL, an RDF query language (Zhang and Beetz, 2016).

Model enrichment is another effort that aims to make a BIM model more readily processable for specific applications such as compliance checking by automatically adding required objects and attributes (Belsky et al., 2016; Farias et al., 2014; Werbrouck et al., 2020). There are also suggested approaches to embed rules into the building model (Hjelseth, 2012; Nisbet et al., 2012; Solihin and Eastman, 2016; Zhang et al., 2013), to enhance object mapping.

Another challenging aspect of BIM data needed for compliance checking processing is spatial operations and computational geometry. There are several important research developments in this area such as BERA (Lee et al., 2015) and Spatial Query Languages (Borrmann and Rank, 2009), (Doherty et al., 2012). Recent work combines building model enrichment using spatial artefacts with logic programming using Answer Set Programming (ASP) to resolve compliance with qualitative normative provisions (Li et al., 2020).

2.2.2. Modelling and accessing normative knowledge

Knowledge acquisition and knowledge engineering are two activities that are central to the modelling and representation of normative provisions as computable rules (Drogemuller et al., 2000; Lee, 2011a; Turk, 1999). Normative knowledge comes in the form of natural language text paragraphs, tabulated and illustrated provisions, prescribed calculation methods, and normative commentaries. Different forms of normative knowledge call for slightly different approaches to represent them. A popular technique of marking up paper-based documents to assist with knowledge acquisition is the RASE method (Hjelseth and Nisbet, 2011), which was previously employed by SMARTCodes (See, 2008).

Conventionally, rules are embedded directly into computer instructions as part of the compliance checking system. Alternatively, the language-based approach has been used to represent normative knowledge as computable rules either in a domain-specific language (DSL) such as BERA (Lee, 2011b), BIMRL (Solihin and Eastman, 2015), VCCL (Preidel and Borrmann, 2015), and RKQL (Dimyadi et al., 2016b), or one of many logic programming languages (Solihin et al., 2019), (Eastman

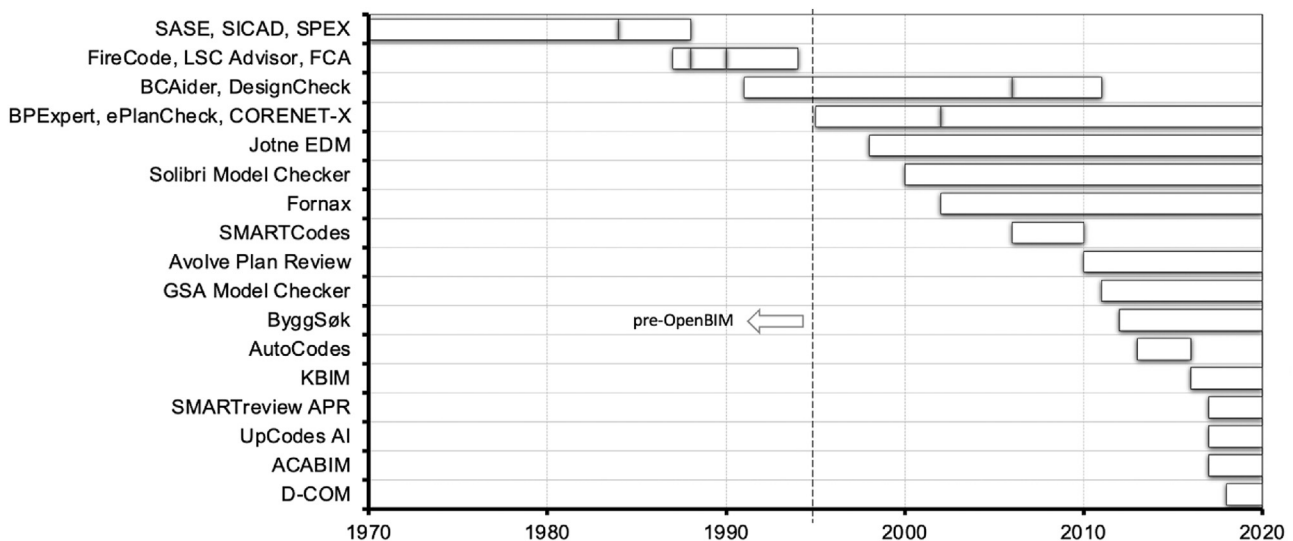


Fig. 4. Timeline of major automated compliance checking systems (ACCS).

et al., 2009).

In the legal domain, research on sharing of legal information also dates back as early as 1960 when computerised services and document retrieval systems were first implemented in the US and UK (Bourne and Hahn, 2003). The most recent and prominent effort in representing legal knowledge is the standardisation of Akoma Ntoso by OASIS into LegalDocML, which is intended to represent the literal content of a legal document (Dimyadi et al., 2017a), (Vitali and Zeni, 2007). A parallel project that is intended to represent the semantic or logical content of a legal document is LegalRuleML, which is also undertaken by OASIS (Palmirani et al., 2011). Similar approaches that have been reported in the literature include LKIF (Gordon, 2008) and CEN MetaLEX (Boer et al., 2008).

The NLP (Natural Language Processing) approach to capture normative knowledge from natural language normative provisions has been gaining traction over the years (Van Gog and Van Engers, 2001; Voorhees, 1999; Zhang and El-Gohary, 2016). This is an area of research that has started to show real potential in the automated development of computable rules from their natural language sources.

The D-COM project led by the University of Cardiff in the UK sets out to push for the adoption of the digitization of regulations, requirements and compliance checking systems in the built environment by 2025 (CDBB, 2019).

2.2.3. Compliance checking methods and systems

There are many variations to the rule-based approach to compliance checking (Beach et al., 2015; Dimyadi et al., 2016c; Eastman et al., 2009; Farias et al., 2014). One of the earliest large-scale ACCS is the CORENET's ePlanCheck (Khemplani, 2005), commissioned by the Singapore government, although it has not been fully utilised in practice. Recently, Singapore's government agencies, BCA, URA, and PUB are collaborating with local vendors to revisit the project with the aim to deliver the next generation of ACCS called CORENET-X. Another recent large scale ACCS is KBIM, initiated by the South Korean government in collaboration with buildingSMART Korea and several research institutions there (Lee et al., 2016b). The objective is to enable checking of compliance with the Korean Building Act, which is the governing legislation for all construction work in South Korea. KBIM makes use of KbimCode, the computable rules representation of their normative provisions.

2.3. Recent ACCS developments

2.3.1. Add-on to BIM authoring tool

Several commercial ACCS have been released to serve the industry in recent years. This includes SMARTreview™ APR™, which is an add-on to Autodesk's Revit BIM authoring tool that supports portions of the International Building Code (Clayton, 2013). Another commercial Revit add-on ACCS is UpCodes AI (Beach et al., 2020), which also supports portions of the International Building Code as well as several other codes for different jurisdictions across the US. These add-ons provide designers with compliance advice within the Revit software environment.

2.3.2. Human-guided automation approach to compliance checking

Another recent commercial development is ACABIM, which is the result of research in New Zealand to find a more pragmatic approach to ACC. ACABIM has been used in a pilot BIM-enabled consenting project as well as a case study by a Building Consent Authority (BCA) in New Zealand (Dimyadi and Amor, 2018). The philosophy behind this stand-alone ACCS is a human-guided workflow-driven approach to automating computable compliance tasks that are tedious, so that human experts can focus on checking performance-based designs that are more qualitative in nature and that requires some tacit knowledge to perform, which machines are not suited to undertake. ACABIM fully supports open standards (Fig. 5) and utilises OpenBIM to share building information and LegalRuleML to represent normative knowledge. The human-guided ACABIM workflow employs a DSL that can query data from various

sources and perform calculations on the data. The benefit of this 'white-box' human-guided approach is that the mapping of objects and attributes is specified and predefined by human experts in the form of executable workflows that can be chained and saved for repeated compliance checking processing (Dimyadi et al., 2017b). This, effectively, provides a means of automating the execution of checklists and flowcharts that BCA and industry's practitioners are already using in manual compliance checking processes. The approach also supports supplementary human input as well as interaction with simulation processes that augment the compliance checking processes.

Like many other countries (Gen et al., 2015), (Wix and Espedokken, 2004), the New Zealand government has also provided the support needed to computerise several priority documents within the New Zealand Building Code in LegalRuleML, an open initiative that tools such as ACABIM can take advantage of (Dimyadi et al., 2020).

2.3.3. Artificial intelligence approach to compliance checking

Research into the use of artificial intelligence (AI) in ACC dates back to the late 1980's and throughout the 1990's (Niwa, 1989), (Adeli, 1991) and has started to gain popularity again in recent years (Verheij, 2020), (Behboudi et al., 2012). Some of these systems attempt to revisit the notion of the distributed expert systems of the early 1990's to provide ad-hoc compliant design advice.

Natural Language Processing (NLP) and Machine Learning (ML) techniques have also been investigated by researchers as a means to capture normative knowledge from paper-based normative provisions in codes and standards (Zhang and El-Gohary, 2012), (Salama and El-Gohary, 2011). Additionally, a well-designed ontology based on AI approaches allows powerful reasoning in performing object and attribute mapping or applying rules to common objects for compliance checking purposes (Li et al., 2020), (Zhang and El-Gohary, 2015).

3. Uncertain present versus optimistic future

As researchers across the world build upon the insights gained from over 50 years of research into ACC, there are a number of core issues where consensus is evident. There are also core unresolved issues that are being consistently identified.

3.1. Core issues with consensus from the research community

3.1.1. BIM contains sufficient information

The open standard representations for buildings (e.g., IFC) have matured into a comprehensive representation sufficient for many core processes in the industry. Researchers are able to use these representations to extract the information required for ACCS and identify structures and relationships in the building as required by code specifications. While national codes have varying information needs the extensible properties of open BIM representations (e.g., PropertySets in IFC) ensure that any extra information requirement can be represented in the BIM model and can be exported to a checking system. The buildingSMART Data Dictionary (ISO 12006-3, 2007) developments give an open specification of concepts in the domain, with equivalences in all languages, to map concepts in codes and standards to those in BIM.

3.1.2. Knowledge representation approaches are sufficient for prescriptive codes

While there is no consensus on a single knowledge representation approach for codes and standards, the various representations in use are proving capable of completely capturing prescriptive codes and standards. The representational capabilities of languages spanning OWL (W3C, 2012) through to LegalRuleML (Palmirani et al., 2011) (and its associated LegalDocML for the layout of legal documents) have been able to be applied to complete codes across several national code sets. Researchers have also identified approaches to represent performance-based codes in their representational languages, but not to

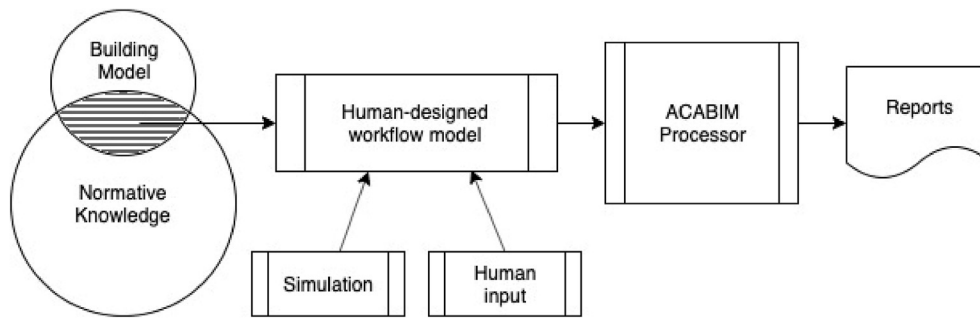


Fig. 5. The human-guided ACC approach of ACABIM.

the extent that automated checking can be applied from the representation.

3.1.3. Solvers can check prescriptive codes

The evidence from a number of research projects and commercialization activities internationally is that with the combination of BIM models and a representation of a prescriptive code it is possible to run a complete compliance check. Checking all possible building types and forms is not possible, but the current constraints on types of building which can be checked allow for the majority of designs to be validated.

3.1.4. Performance-based codes require human expert oversight

Checking approaches for performance-based codes are less amenable to full automation and typically require either the intervention of human experts to make assessments or to utilize simulation tools to gain insight into a building's performance. Many performance-based codes do have a prescriptive pathway (e.g., deemed-to-satisfy or acceptable solutions) that are heavily used by the industry and hence provide for automated checking for many building types. However, human-guided white-box approaches such as that taken by ACABIM (Dimyadi and Amor, 2018), (Dimyadi et al., 2017b), (Dimyadi and Amor, 2019) provide a better means of achieving this that also adapt well to conventional practices.

3.2. Ensuring BIM model quality

The first of the unresolved issues is that the quality of the data in the BIM model to be checked needs to be uniformly high to ensure ACC is able to be undertaken with confidence. Researchers working in the field have identified a range of issues where this is not the case and proposed solutions which aim to ameliorate the issues. These issues are canvassed in the following subsections.

3.2.1. Sufficiency of information

ACC is reliant upon having the necessary information about a building and its components so that checks can be applied automatically. This places a significant burden upon the designer and BIM modelers to ensure that the required information is sourced and included in the model. Across a complete BIM for any reasonably sized building, this is an onerous task and one that BIM tools are not well-designed to support. Any missing information is a significant issue for ACC and will usually result in a halt to the checking process and a request for information (RFI) being generated back to the project team. As building consenting authorities (BCA) are faced with BIM models that typically are not complete, a range of approaches to address this issue are being investigated.

Consenting authorities can look to prime the marketplace by publishing BIM standards for the models they will be accepting (Christchurch City Council, 2014). This can cover the attributes they expect for every type of object, classification systems which will be needed, and expected approaches to modelling. To support these educational approaches for the industry templates can also be developed (e.g., for Revit) which are populated with the required attributes for each object for ACCS.

Standards such as IFC allow for an MVD (Model View Definition) (Building SMART, 2016), which defines the exact subset of entities and attributes required for a process such as ACC. With a defined MVD it is then possible for designers and a BCA to check whether all the required information is within the BIM model prior to it being submitted.

3.2.2. Correct and consistent classification of information

A significant portion of code clauses rely upon a classification assigned to a component, to uniquely signify the type of information present, for example, the room's use function. While there are over 70 major classification systems in existence, a consenting authority will expect components in the BIM model to be classified using one, or perhaps two, nationally acceptable classification systems. Missing classification codes, or the use of a classification system outside of those accepted by the consenting authority will halt the ACCS process. Approaches such as the MVD for IFC can ensure that a classification code is supplied when necessary, but more sophisticated model checkers (e.g. Solibri Model Checker) are required to check whether the specified code belongs to the allowable classification systems.

A more insidious problem with classifications is where a component has been misclassified. ACCS will use a specified classification code to determine relevant clauses to apply, and hence a misclassification can have serious consequences. For example, if the activity of a space is misclassified as a kitchen rather than a corridor. Experience with BIM models shows that this is a not uncommon occurrence. Checking that a classification code is correct for a space or component is a complex problem requiring a sophisticated understanding of buildings and their components. AI approaches are looked at in this regard to help ensure classification codes are correct. Recent work on semantic enrichment (Belsky et al., 2016) shows that ML, or rule-based checking, can be applied with high accuracy to some classification processes (e.g., the activity or function to assign to a space). However, there are many types of classification that have not been addressed by these approaches to date and which will need to be investigated to provide a full coverage of checks for spaces and components of buildings.

3.2.3. Variable approaches to modelling with BIM

There is a plethora of approaches to using BIM tools to model a building. Many modelers come from a 2D and 3D CAD background and bring across approaches from those fields to their BIM modelling. Tertiary institutions also take different approaches to teaching modelling in BIM as do the various BIM companies. Consequently, BIM models can have widely varying representations for similar constructs. A typical example being the use of slab elements to represent many constructs in a BIM model, such as stairs. These approaches to modelling in BIM also cause difficulties for ACCS that expect to find objects of a certain type in the BIM model for their checking (e.g., a stair object for egress rather than dozens of slab objects representing a stairway). BIM modelling guidelines from consenting authorities can give guidance as to how BIM models are expected to be constructed, but if a different approach is taken then new methods of checking a model are required. A semi-automated

human-guided approach of compliance checking is a pragmatic interim solution. BIM-enrichment techniques also provide pre-processing support to ACCS. AI approaches have the potential to identify common modelling practices, such as slabs being used to represent stairways, or misalignment of structural elements, or unconnected elements. However, automated correction of BIM models through AI approaches is still an area of research and there are a huge number of potential checks that may need to be undertaken in a BIM model.

3.2.4. Cost to create a complete BIM model

Where a consenting authority wishes to make automated checks on a BIM model there is a shift of effort required in the industry to supply the information required for the checks. Unlike the existing approach to code compliance checking all information must now reside in the BIM model in the appropriate objects, attributes and property sets. The BIM model also needs to be of a uniformly high standard to enable the various rule checks and simulations that might be run for ACCS. This is often beyond the current expectations of BIM modelers and design teams, meaning extra effort is put into the creation of the BIM model for consenting, even if the information required is no different from existing processes. A range of approaches can be considered for this issue. Different forms of contract could be utilised when ACCS is in place, to ensure the extra modelling effort is recognised and compensated. Greater building knowledge could be imbued in BIM tools to ensure that common errors are impossible to generate in a BIM model (e.g., walls not flush with floor slabs), and company specific policies and design constraints could be used to help generate information in the evolving BIM model.

3.2.5. Correct generation of BIM data for ACCS

A constant issue for BIM tools is the correct generation of data in the interoperability format that will be used for ACCS (e.g., translation to IFC). It is well known that perfectly correct software is impossible to develop for any system with more than a few simple functions. The software industry therefore has a range of techniques to try to give better certainty to the quality of a software product, but in the knowledge that there will always be bugs. For ACCS this is a significant issue as BIM models, even from certified BIM tools, can contain erroneous data from the translation processes which are used to support the interoperability required for ACCS. While it was possible to undertake some checking of a translated building when using 2D and 3D CAD (e.g., by eye-balling the generated graphical representation) this is not as effective for BIM as the majority of the transferred data is in attributes and property sets which have no geometric form. In the future this may be addressed by more comprehensive certification of BIM tools, especially against ACCS requirements. It is also an issue where AI approaches may provide further assistance by checking for anomalous information in the generated transfer format for ACCS.

3.3. Quality and transparency of translated codes

The creation of computer interpretable versions of codes and standards has the potential for significant improvement in the quality of the codes. Early work on computerisation of codes identified the benefits in ensuring consistency of terminology across a nation's codes (Vanier et al., 1994) as well as removing inconsistencies that exist in clauses across independently developed codes (Vanier et al., 1994). The harmonisation of terms across a nation's codes is an initiative that aligns well with developments such as the bSDD (ISO 12006-3, 2007). This will also make a nation's codes more consistent and coherent for all those looking to use them.

The creation of computer interpretable versions of codes and standards comes at a high cost when this is being undertaken by human experts. Experience in New Zealand has shown that it takes approximately a day of an expert's time to translate a page of a code and undertake the quality control processes on the translation. Multiply this by the hundreds of codes and standards available in each nation (e.g.,

around 600 available for use in New Zealand), some of which have hundreds of pages, and this quickly adds up to several millions of dollars for the conversion. The human translators also need specialised expertise spanning Architecture, Engineering and Construction (A/E/C) as well as the computing logic utilised in the representation it is translated into. There are very few people with this combination of skills currently and it is a capability which would need to be grown to undertake significant conversion of a nation's codes and standards.

Codes and standards, being subsidiary legislations, are frequently amended as construction approaches change and new technologies come onto the market. So, maintaining computerised versions alongside the paper-based versions is going to be a major cost for any nation wishing to move down this path. To tackle this issue there is a strong interest in the application of NLP to the text of codes and standards (Zhang and El-Gohary, 2016). This has the potential of creating systems which understand the intent of codes and standards as well as automating the translation into a computer interpretable format.

As paper-based codes and standards are published openly, typically by standards organisations within a country, it is possible for anyone to understand what is required by a code or standard. This becomes more difficult when a computer interpretable version is created, though it is vital to be able to check the equivalence of the paper-based code and the computerised analogue. In particular, black-box representations of a code or standard are not amenable to checking and validation by the user of a system which incorporates that representation. Unfortunately, that is the dominant mode of development of computerised versions for commercial systems to date (Solibri Inc, 2020) (UpCodes, 2020). With a range of strong representation languages which would allow open publication of a code or standard (e.g., OWL/SWRL or LegalRuleML), there is a strong case to argue for standards organisations and governments to develop computerised analogues of their paper-based codes and standards and to distribute them alongside their paper-based versions.

While prescriptive codes and standards are amenable to computerised representation it is more difficult to fully encode a performance-based code or standard. Further development of languages such as LegalRuleML will be needed to enable a representation of performance aspects of codes and standards and to allow checkers to find a pathway to ensuring compliance with these types of clauses.

3.3.1. Robust quality assurance and quality control processes

Quality assurance and quality control is a vital part of ACC and needs to be applied in several parts of the process. When codes and standards are being translated into a computer interpretable format there needs to be a guarantee that they are equivalent to the original paper-based versions. Currently, there are no standard and recognised processes to achieve this. Across the world a range of approaches is used for quality assurance, ranging from having a hierarchy of oversight and checking of the translation, to training experts in a controlled translation approach (Peng and El-Gohary, 2015). Experience in code translation shows that when two different experts are asked to translate the same clauses, even with extensive training, there will be differences in the resultant translation. However, this is seen as best-practice to developing a gold-standard, giving the ability to measure the inter-annotator agreement and to have a third party to adjudicate discrepancies (Pestian et al., 2012). Where researchers turn to computerised approaches, for example through the use of NLP and other AI techniques, there are still significant variations to the expert translations. When the AI techniques use a learning component there is an added complexity that the results are not deterministic, and with time and increased learning by the NLP or AI system the translations will change. This is obviously not a good trait for a translation system to exhibit.

Alongside the quality assurance there needs to be a process of quality control, to detect when incorrect translations are made. Researchers have trialled regeneration of human readable clauses from the translated code as one way of being able to determine the quality of a translation. However, the gap between the written form of the generated clauses and

the original documents is high, making this a poor quality control approach. Oversight by expert checkers is often utilised for translations, but also doesn't give a perfect result in regard to the quality of the final translation.

The use of DSL (Domain Specific Language) has been trialled by a number of groups internationally to help with quality assurance (Preidel et al., 2018). A DSL provides a higher-level specification of a code or standard which can be used to automatically generate the detailed representations such as OWL and LegalRuleML. The DSL often provides a more structured approach to translation of codes and is easier to read than the lower level representational languages that a computer will need to check a code (Dimyadi et al., 2020). Further development of DSL and their specification environments look to be a core part of future code translation approaches.

Finally, the quality control approach for ACCS is also an area of considerable development. As with many other domains that computerised approaches are being developed in, there is a need for a significant database of gold-standard test cases where the ground truth translation has been established. Trials against a small number of clauses has been undertaken internationally by buildingSMART in their regulatory room activities (buildingSMART, 2017), but there is no established database of clauses and translations that can be used to test the quality of ACC which has good coverage of the wide range of possible checks which need to be made by an ACCS.

3.3.2. Extending checker functionality

The different ACC checkers which have been developed to date have taken significantly different approaches to the underlying engine which powers their system. This leads to systems which exhibit different strengths in their capabilities to check particular types of clauses. Some checkers have a strong spatial bias which enables a wide range of sophisticated spatial predicates to be represented and checked (Borrmann and Rank, 2009). For example, adjacencies, or notions of above and below, for arbitrarily shaped spaces. Another approach is to utilize logic tools such as Answer Set Programming (ASP) with spatial artefacts to ascertain the result of complicated clauses (Li et al., 2020). For example, to determine whether a sanitary fixture is visible from an adjacent circulation space. What these approaches highlight is that there is no complete understanding of what functions are required to support a complete ACC. As more codes are examined and national systems developed, it is expected that this understanding will grow. One potential outcome of this understanding would be of a library of ACC functions which can be utilised in any effort to develop an ACCS, providing assurance that there would be sufficient functional coverage to enable the majority of clauses to be checked.

One aspect of an ACCS that has not received significant attention is how the checking process is orchestrated. The process is typically not evident in the codes and standards but often determined by the appropriate BCA. While translated code clauses can be applied individually to elements within a BIM model, this does not give an orchestration which can be seen to be equivalent to the process undertaken by BCAs. Some ACCS look to encode the process that a BCA follows currently to provide an obvious equivalence in approach to help in the quality control of the ACCS in relation to a human expert checking process (Dimyadi and Amor, 2018). As this approach of having an executable checking flow-chart is well aligned to matching ground truth exemplars in a test case database, it is expected that a greater number of ACCS will provide a process notation to direct the checking process. As most process notations have a visual aspect (e.g., BPMN), this provides a representation which is easily understandable by human experts looking to verify that the interpretation of clauses in a code is being undertaken as in the currently accepted manual process.

3.3.3. Scaling to a national level

Scaling ACC to a national level requires the development of large-scale systems and frameworks which are significantly different from

those suitable for research projects into particular aspects of an ACC. South Korea and Singapore, who have committed to the development of national ACC for their codes and standards, have instigated major development programmes with KBIM and CORENET-X respectively. Systems which need to be developed when scaling up include:

- **Code and standard servers.** With approximately 600 codes and standards available in a nation there needs to be an effective method for distribution of the latest computable version of a code or standard. The development of code servers allows an investigation of models at the quantum at which code information is served. Options ranging from a whole code or standard through to individual clauses, equations, tables, etc. It also supports intelligent identification of clauses applicable to a particular construct in a building across the whole code and standard set.
- **Publication of the consenting model(s).** The information requirement for each type of consent check can be published openly, for example using MVDs for the IFC schema. This would allow checkers to be developed to ascertain whether the required information is present for the ACC process prior to submitting the BIM model. This could be supported by templates for the major BIM software tools to ensure that the required consenting information needs are visible during design and detailing.
- **Training and best practice for modelling.** The industry would need to be supported with training for best practice in modelling for consenting. Along with design guidelines the BCAs would need to transition the industry to providing BIM models in a particular form.
- **National BIM Object Library.** The development of a national resource for BIM objects would be beneficial for a nation to ensure that required information is already attached to the objects that would be used to develop a BIM model. Countries like the UK are already well on the path to establishing a national library containing both generic object representations and those from specific manufacturers (NBS, 2020).
- **National checking procedures.** In most countries, the checking procedures used by each BCA are designed independently. Leading to situations where national codes and standards have different interpretations in different parts of a country. Scaling to a national ACCS provides an opportunity to harmonise the checking process for each of the codes and standards.
- **Semantic enrichment tools for the industry.** Accepting that BIM models will not be perfect, a national ACCS provides an opportunity to develop support tools for the industry to help validate, rectify and complete BIM models for ACC. As well as supporting ACC, such semantic enrichment approaches will significantly improve the quality of BIM models which are used in projects and hence the trust and uptake of BIM by the industry.

4. Conclusions

In the last half century, incredible research effort has been applied to progress the field of Automated Compliance Checking (ACC). The understandings generated from these efforts are significant, leading the community to a point where commercial systems able to provide checking against a nation's codes and standards are being realised. This has been achieved by three parallel fields of research maturing sufficiently to support the sophisticated needs of ACC. The development of BIM, and especially standards-based OpenBIM (e.g., IFC), ensures that all the building information required for ACC can be sourced. The development of knowledge representation standards, and especially open standards (e.g., OWL and LegalRuleML), ensure that the computable content of codes and standards can be represented. The development of process enacted solvers, especially connected with open process notations (e.g., BPMN) and sophisticated building-aware query languages (e.g., BIMQL or QL4BIM), enable an orchestration of BIM and computable codes and standards to check complete buildings against

prescriptive specifications.

Despite the prestigious progress made, there are still a number of areas which require further research development. The quality and completeness of BIM models has been found wanting and it is clear that achieving the required quality threshold implies a major change in the industry, perhaps a generational change. To enable progress in this decade, research in enrichment and checking of BIM models will be necessary. The human inability to consistently translate information impacts our ability to ensure exact translations of codes and standards. NLP and AI approaches to understanding textual documents provide a pathway to gaining gold-standard translations. Quality assurance and quality control still need to be developed for our knowledge representations as well as a significant database of gold-standard translations that researchers can test their systems upon, as exist in many other computing domains. The evolving solvers need to become more sophisticated, with support for the many important concepts found in buildings, as well as the needs of performance-based codes which are not sufficiently supported currently. Finally, as ACC is scaled to a national level there are a range of support systems which would ease their progress. These range from sophisticated computable code servers, BIM object libraries, standardised checking processes, add-ins for BIM tools to support correct model generation, etc.

As has been evident for the last half century, the field of ACC is vibrant and attracts considerable research attention. There are many challenges still to consider which will keep researchers engaged for at least the next half century.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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